

Uncertainty-Aware Cross-Modal Opinion Interaction: A General Framework for Visible-Infrared Vehicle and Person Re-Identification

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Abstract

Visible and Infrared Re-Identification (VI-ReID) bridges the semantic gap between visible and infrared heterogeneous data, supporting all-weather intelligent surveillance. Existing methods are limited by static modal interactions and insufficient uncertainty awareness in complex scenarios. To solve this, we propose a unified Cross-Modal Opinion Interaction network (CMOI) for both vehicle and person cross-modal ReID, consisting of three novel modules: modal opinion generation, cross-modal opinion correlation, and joint opinion optimization. CMOI converts deterministic feature matching into a probabilistic decision paradigm by quantifying cross-modal prediction uncertainty, defined as model "opinion". It then dynamically exchanges uncertainty-aware predictions to form joint opinions, correcting low-confidence predictions and enabling modal complementary learning. In optimization, cross-entropy and regularized KL loss are adopted to evaluate prediction accuracy and improve generalization on uncertain samples. Extensive experiments on five visible-infrared vehicle and person datasets validate its effectiveness. Notably, CMOI improves Rank-1 accuracy by 6.32% on DN348, achieving highly competitive performance.

1. Introduction

Person re-identification (ReID) [20] and vehicle ReID [26] are pivotal tasks in computer vision, aiming to recognize the same object across different camera views. Although deep learning has propelled significant progress in single-modal (visible) ReID, it struggles under poor lighting conditions such as night or haze. Infrared (IR) imaging, rely-

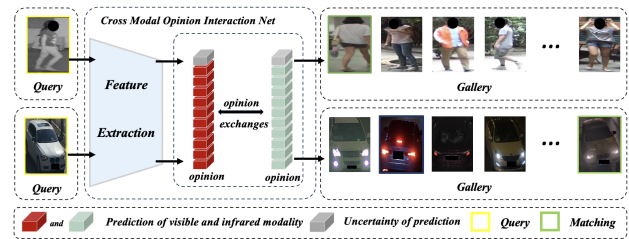


Figure 1. Illustration of the proposed Cross-Modal Opinion Interaction (CMOI) network, which is applicable to both visible and infrared re-identification for vehicles and persons.

ing on thermal radiation, complements visible imaging for continuous surveillance. Visible-infrared person and vehicle re-identification (VI-ReID) [29] has emerged as a solution, integrating visible and infrared data. This cross-modal approach enables robust object recognition under diverse illumination, bridging the gap left by single-modal methods [12].

Compared with images in single-modal re-identification tasks [21], visible and infrared images exhibit significant modal heterogeneity in spectral response characteristics, texture details, and feature space distribution. This inherent modal difference makes visible-infrared person and vehicle re-identification far more challenging. Current mainstream solutions fall into two paradigms: feature-level alignment and image-level transformation [1]. The former [30, 32] normalizes cross-modal feature distributions by constructing shared feature space mappings but tends to enforce homogeneous feature representations, inadvertently suppressing modality-specific discriminative information. The latter [6, 24] leverages generative models to synthesize auxiliary modal data or augment training samples, yet suffers from distortion in synthetic details and over-reliance on generative quality. Both paradigms share a critical limitation:

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they treat modal interactions as static, deterministic processes, failing to quantify prediction uncertainty or dynamically exploit cross-modal complementarity based on confidence levels. Moreover, existing methods often focus exclusively on either person or vehicle ReID, lacking a unified framework to address the common challenges of cross-modal recognition across object types.

To overcome these limitations, we propose a Cross-Modal Opinion Interaction (CMOI) network, which innovatively transforms the static modal alignment paradigm into a dynamic, uncertainty-aware collaboration mechanism through a three-stage architecture. This unified framework addresses cross-modal person and vehicle ReID by modeling modal interactions as “opinion exchanges” grounded in probabilistic uncertainty quantification, as illustrated in Figure 1.

Modal Opinion Generation: Traditional VI-ReID methods output deterministic feature embeddings, ignoring prediction uncertainty and risking erroneous matches in ambiguous scenarios. CMOI introduces a Dirichlet distribution modeling framework [22] to upgrade deterministic matching to probabilistic decision-making. For input visible and infrared data, after feature extraction, key embedding information is mapped to the parameters of the Dirichlet distribution via a dedicated function. This distribution systematically characterizes epistemic uncertainty in cross-modal predictions, with its parameters intuitively representing the model’s confidence in different category predictions—forming the model’s “opinion” [13]. Unlike feature-level alignment methods that homogenize modalities [30, 32], this approach preserves modality-specific characteristics while quantifying uncertainty, providing a confidence basis for subsequent decision-making. For example, visible modality opinions emphasize confidence in color and texture, while infrared opinions highlight reliability in thermal contours, laying the groundwork for targeted interaction.

Cross-Modal Opinion Correlation: Existing image-level transformation methods [15, 35] unilaterally synthesize modal data without considering dynamic complementarity. In contrast, CMOI implements a bidirectional dynamic interaction mechanism that leverages multimodal complementarity based on opinion confidence. For low-confidence predictions in one modality (e.g., visible images), the network retrieves high-confidence cues from the other modality (e.g., infrared contours) for correction. This interaction is not merely feature fusion [24, 30] but a targeted “opinion negotiation”: visible opinions guide infrared focus on texture-correlated thermal regions, while infrared opinions refine visible judgments in low-light areas. This process generates a joint opinion that integrates the strengths of both modalities, addressing the static nature of existing alignment strategies [9].

Joint Opinion Optimization: To balance prediction accuracy and uncertainty awareness, CMOI employs a Bayesian decision framework with Dirichlet priors. It optimizes two complementary losses: cross-entropy loss ensures prediction accuracy by aligning joint opinions with ground-truth categories, and regularized uncertainty loss mitigates overfitting to uncertain samples. This dual-loss strategy surpasses traditional metric learning [21] by explicitly controlling generalization on ambiguous data, ensuring robust performance across diverse scenarios—from well-lit person and vehicle datasets to nighttime surveillance.

Our key contributions are:

- **Paradigm Shift:** By mapping evidence to Dirichlet distributions, CMOI quantifies cross-modal uncertainty and outputs “model opinions” with confidence scores, supplementing deterministic feature matching with an interpretable, probabilistic decision layer.
- **Dynamic Complementarity:** Unlike static alignment [28] or one-way synthesis [24], CMOI’s interaction mechanism dynamically corrects low-confidence predictions using cross-modal cues, maximizing the synergy between modalities.
- **Unified Framework:** CMOI is applicable to both visible-infrared vehicle and person re-identification, outperforming specialized methods on five benchmark datasets.

2. Related work

2.1. Visible and Infrared vehicle Re-identification

In the field of Visible and Infrared Vehicle Re-identification (VI-Vehicle ReID), existing research has explored both single-modal and cross-modal directions. In single-modal vehicle re-identification, VAT [41] adopted a viewpoint-aware attentive multi-view inference model to acquire multi-view vehicle features, enhancing the model’s adaptability to different observation angles; PRND [10] integrated a detection branch to learn part-regularized features, focusing on key vehicle components to improve recognition accuracy. In the visible-infrared cross-modal scenario, existing methods face severe challenges from modal differences. DNDM [16] introduced a dual-domain structure enhancement module for visible-infrared cross-domain vehicle re-identification, extracting local pixel-level gradient features to strengthen the structural consistency of vehicles in low-light environments (such as body contours and head-light layouts).

2.2. Visible and Infrared Person Re-identification

In the field of Visible and Infrared Person Re-identification (VI-Person ReID), existing methods are mainly categorized into image-level and feature-level approaches. Image-level methods typically employ Generative Adversarial Networks (GANs) to convert infrared person images into vis-

ible images and vice versa, aiming to reduce modal discrepancies. For instance, some studies [2, 5] introduced identity-preserving constraints to ensure that person identity information remains intact during image conversion. However, such methods are often affected by variations in person poses and clothing textures, leading to artifacts in generated images that affect subsequent recognition performance. Additionally, intermediate-modality conversion methods [35, 40] can alleviate modal differences to some extent but impose additional training data requirements. Meanwhile, semantic gaps persist between intermediate representations and original modalities. Feature-level methods [3, 8] based on attention mechanisms can enhance features of key person parts, yet they still face challenges of insufficient modal fusion in complex scenarios.

3. Method

3.1. Problem Formulation

Let $\mathcal{X}_{vis} = \{x_{vis}^i\}_{i=1}^N$ and $\mathcal{X}_{ir} = \{x_{ir}^i\}_{i=1}^N$ denote visible and infrared image sets, respectively, where each paired sample (x_{vis}^i, x_{ir}^i) shares the same semantic label $y^i \in \{1, 2, \dots, K\}$, with K representing the total number of categories. ResNet-50 [11] extracts cross-modal shared features $\mathbf{f}_{vis}^i, \mathbf{f}_{ir}^i \in \mathbb{R}^D$ from x_{vis}^i and x_{ir}^i , respectively. The network outputs a predicted probability distribution $\hat{y}^i$ over K classes for each sample pair.

To optimize the cross-modal shared features, we define two baseline losses: the identity loss ($\mathcal{L}_{id}$), which aligns predictions with labels, and the triplet loss ($\mathcal{L}_{tri}$), which enforces intra-class feature similarity and inter-class dissimilarity. These two loss functions are defined as:

$$\mathcal{L}_{id} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \mathbb{I}(y^i = k) \log \hat{y}_k^i, \quad (1)$$

$$\mathcal{L}_{tri} = \frac{1}{N} \sum_{i=1}^N \max(|\mathbf{f}_{vis}^i - \mathbf{f}_{ir}^n|^2 - |\mathbf{f}_{vis}^i - \mathbf{f}_{ir}^p|^2 + p, 0). \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function. And $\mathbf{f}_{ir}^p$ and $\mathbf{f}_{ir}^n$ are positive (same class) and negative (different class) infrared sample features, respectively, with p as a predefined margin.

3.2. Modal Opinion Generation

To quantify prediction uncertainty for visible and infrared cross-modal data, feature-to-evidence transformation is first performed. For input visible modal data x_{vis}^i and infrared modal data x_{ir}^i , feature vectors $\mathbf{f}_{vis}^i$ and $\mathbf{f}_{ir}^i$ are extracted via ResNet-50 feature extractors, then converted into evidence vectors $\mathbf{e}_{vis}^i \in \mathbb{R}^K$ and $\mathbf{e}_{ir}^i \in \mathbb{R}^K$ through Fully Connected (FC) layers:

$$\mathbf{e}_{vis}^i = \text{ReLU}(\text{FC}(\mathbf{f}_{vis}^i)), \quad \mathbf{e}_{ir}^i = \text{ReLU}(\text{FC}(\mathbf{f}_{ir}^i)). \quad (3)$$

where $e_{vis}^{i,k}$ and $e_{ir}^{i,k}$ denote the evidence amount for category k in the i -th sample under the visible and infrared modalities, respectively. The FC layer aligns the feature vector with the dimension of the training class space (i.e., K classes), while the ReLU function introduces non-linear transformation and ensures non-negativity of the output. This process provides an interpretable semantic basis for uncertainty quantification, as evidence values directly reflect the intensity of support from each modality for different categories.

Subsequently, Dirichlet distribution parameterization is conducted by mapping evidence vectors to Dirichlet distribution parameters $\alpha: \alpha_{vis}^k = e_{vis}^k + 1, \alpha_{ir}^k = e_{ir}^k + 1$. For uncertainty quantification metrics, the belief mass b_k explicitly quantifies category confidence:

$$b_k = \frac{\alpha_k - 1}{S}, \quad S = \sum_{k=1}^K \alpha_k, \quad u = \frac{K}{S}. \quad (4)$$

where S is the Dirichlet strength reflecting the total evidence amount. b_k normalizes the model's belief degree that the sample belongs to category k to $[0, 1]$, making confidences across modalities comparable. The overall uncertainty u reflects the model's "unknown degree" of prediction.

The softmax function [33], serving as a classification output layer, calculates a deterministic probability distribution vector b whose elements sum to 1. It can only represent the classification probabilities of samples but ignores the uncertainty of predictions. In contrast, the Dirichlet distribution describes the probability distribution and quantifies the uncertainty of the probability distribution simultaneously via the parameter α (a positive real-valued vector with the same length as the number of classes).

Take a three-class classification task as an example. As shown in Figure 3, when the Dirichlet parameter $\alpha = [41, 2, 2]$, $S = 41 + 2 + 2 = 45$, so $b_1 = (41 - 1)/45 \approx 0.93$ (confidence for the first class) and $u = 3/45 \approx 0.069$ (uncertainty). For comparison, the softmax function might output $b \approx [0.95, 0.02, 0.02]$, which only reflects high probability for the first class. However, the Dirichlet distribution not only indicates high confidence in the first class but also low uncertainty, meaning the model is highly certain with few potential uncertain factors. When $\alpha = [3, 3, 3]$, $S = 9$, so $b_1 = (3 - 1)/9 \approx 0.22$ and $u = 3/9 \approx 0.33$. The softmax function might output $b \approx [0.33, 0.33, 0.33]$, showing balanced probabilities, but the Dirichlet distribution further reveals balanced hesitation: equal α values indicate no bias toward any class, with moderate uncertainty. When $\alpha = [1.01, 1.01, 1.01]$, $S = 3.03$, so $b_1 = (1.01 - 1)/3.03 \approx 0.003$ and $u = 3/3.03 \approx 0.99$. Softmax might still output $b \approx [0.33, 0.33, 0.33]$, but the Dirichlet distribution indicates nearly zero confidence in the first class and extremely high uncertainty.

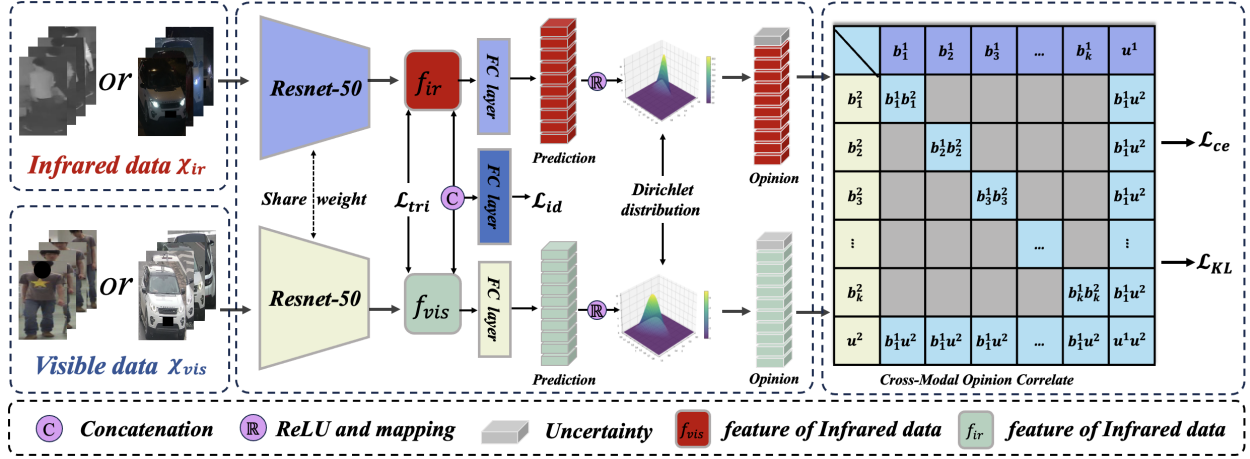


Figure 2. Illustration of the Cross-Modal Opinion Interaction (CMOI) network. Cross-modal sample pairs (x_{vis}^i, x_{ir}^i) with consistent ground truths are fed into ResNet-50 for feature extraction. The CMOI is optimized by the identity loss $\mathcal{L}_{id}$, triplet loss $\mathcal{L}_{tri}$, cross-entropy loss $\mathcal{L}_{ce}$, and KL divergence $\mathcal{L}_{KL}$.

By mapping cross-modal features to the Dirichlet distribution and quantifying uncertainties, this approach realizes a paradigm shift from “deterministic matching” to “probabilistic decision-making” [38]. It not only adapts to the semantic gap between visible and infrared modalities, enhancing model robustness in complex scenarios (e.g., night-time, haze) but also provides quantitative support for cross-modal fusion via evidence theory, enabling complementary integration of modal information.

3.3. Cross-Modal Opinion Correlation

We utilize cross-modal opinion correlation to fuse opinions [9, 22] with uncertainty from different modalities, thereby achieving complementary integration of cross-modal information. The Dirichlet distribution of each modality can be transformed into a “opinion” $\mathbb{M} = \{b_1, b_2, \dots, b_K, u\}$, where b_k represents the belief mass of category k , and u denotes the overall uncertainty. They satisfy the equation:

$$u + \sum_{k=1}^K b_k = 1. \quad (5)$$

For the visible modality opinion $\mathbb{M}^1 = \{b_k^1, u^1\}$ and the infrared modality opinion $\mathbb{M}^2 = \{b_k^2, u^2\}$, the fused joint opinion $\mathbb{M} = \{b_k, u\}$ is calculated as follows:

$$b_k = \frac{1}{1-C} (b_k^1 b_k^2 + b_k^1 u^2 + b_k^2 u^1), \quad u = \frac{1}{1-C} u^1 u^2. \quad (6)$$

where the conflict coefficient $C = \sum_{i \neq j} b_i^1 b_j^2$ measures the degree of conflict between the opinions of the two modalities, and $\frac{1}{1-C}$ serves as the normalization factor. The higher

the conflict coefficient C , the greater the disagreement between the two modalities. Correspondingly, the proportion of uncertainty u in the fused result increases. The combined Dirichlet distribution parameter α_{com} can be derived from the fused joint opinion $\mathbb{M} = \{b_k, u\}$ as follows:

$$\alpha_{com} = b_k \times S + 1 = b_k \times \frac{K}{u} + 1. \quad (7)$$

The fused joint opinion $\mathbb{M} = \{b_k, u\}$ exhibits three key properties:

Property 1: Consistency of Low/High Confidence

When both modalities have low confidence levels, the final classification inevitably yields low confidence. Conversely, if both modalities demonstrate high confidence, the final classification generally yields a high level of reliability. This property ensures that the fused result accurately reflects the overall certainty conveyed by the individual modalities.

Property 2: Dominance of High-Confidence Modalities In cases where only one modality exhibits high confidence while the other shows low confidence, the final classification is predominantly determined by the modality with higher confidence. This mechanism allows the fusion process to prioritize reliable information, effectively leveraging the strengths of individual modalities to enhance prediction robustness.

Property 3: Conflict-induced Uncertainty When two modalities conflict (high C), final classification confidence drops significantly—consistent with evidence theory, where contradictory information reduces fused result reliability. By quantifying and integrating this conflict into $\mathbb{M}$, the framework accounts for modal discrepancies, ensuring predictions reflect true uncertainty. To highlight the advantages

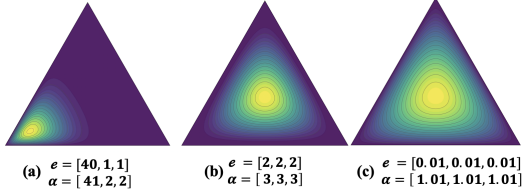


Figure 3. Illustration of the Dirichlet distribution. (a) Unimodal Concentration: When one α parameter is dominant (e.g., $\alpha_i \gg \alpha_{j \neq i}$), the probability density is sharply concentrated at the corresponding vertex of the simplex. (b) Symmetric Concentration: the probability density is concentrated around the center (barycenter) of the simplex. (c) Uniform Distribution: the density is uniformly distributed over the entire simplex.

of our fusion rule, consider two cases. When a blurry visible image (low confidence: $b_{vis} = [0.2, 0.1, 0.3]$, $u_{vis} = 0.4$) pairs with a clear infrared image (high confidence: $b_{ir} = [0.7, 0.1, 0.1]$, $u_{ir} = 0.1$), our rule fuses them into $b = [0.69, 0.09, 0.16]$, $u = 0.06$, where the high-confidence infrared effectively corrects the visible opinion, reducing uncertainty and leveraging modal complementarity. When a visible image favors Category 2 ($b_{vis} = [0.1, 0.4, 0.1]$, $u_{vis} = 0.4$) while infrared favors Category 1 ($b_{ir} = [0.4, 0.1, 0.1]$, $u_{ir} = 0.4$), the fused result becomes $b = [0.29, 0.29, 0.12]$, $u = 0.30$, where increased uncertainty avoids overconfidence in conflicting inputs—unlike static fusion methods [23] that either ignore conflicts or blindly follow one modality, our rule dynamically balances reliability and conflict to ensure robust decisions.

3.4. Joint Opinion Optimization

Traditional cross-entropy loss is limited to optimizing deterministic prediction probabilities (e.g., softmax outputs) and fails to account for uncertain prediction values, such as the belief mass and uncertainty encapsulated in cross-modally fused joint opinions [14]. To address this limitation, we design two complementary loss functions that model uncertainty via the Dirichlet distribution, enabling dual supervision through a probabilistic cross-entropy ($\mathcal{L}_{ce}$) and KL divergence ($\mathcal{L}_{KL}$).

$$\mathcal{L}_{ce} = \mathbb{E}_{\mathbf{b}_y \sim \text{Dir}(\alpha)} [-\log b_y] = \psi(S) - \psi(\alpha). \quad (8)$$

where $b_y \sim \text{Dir}(\alpha)$ denotes the probability that a sample belongs to the true class y under the Dirichlet distribution, with $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_K)$ as the distribution parameters; $\psi(\cdot)$ is the Digamma function; $S = \sum_{k=1}^K \alpha_k$ represents the total evidence strength of the Dirichlet distribution; and α_y is the parameter of the Dirichlet distribution corresponding to the true class y . $\mathcal{L}_{ce}$ measures the degree of matching between the model's predictions (Dirichlet distribution) and the true labels, and is essentially a probabilistic distribution extension of the traditional cross-entropy. The proof of Eq. 8 consists of main steps:

Step 1: Marginal distribution property

The marginal distribution of any component b_y follows a Beta distribution:

$$b_y \sim \text{Beta}(\alpha_y, S - \alpha_y) \quad (9)$$

with probability density function:

$$f(b_y; \alpha_y, S - \alpha_y) = \frac{1}{B(\alpha_y, S - \alpha_y)} b_y^{\alpha_y-1} (1 - b_y)^{S-\alpha_y-1} \quad (10)$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function.

Step 2: Expectation of the log-transform

The expectation of $\log b_y$ under this marginal distribution is:

$$\begin{aligned} \mathbb{E}[\log b_y] &= \int_0^1 \log b_y \cdot f(b_y; \alpha_y, S - \alpha_y) db_y \\ &= \frac{\int_0^1 (\log b_y) b_y^{\alpha_y-1} (1 - b_y)^{S-\alpha_y-1} db_y}{B(\alpha_y, S - \alpha_y)} \end{aligned} \quad (11)$$

Step 3: Relationship to Beta function derivative

The integral can be expressed as a derivative of the Beta function:

$$\begin{aligned} \int_0^1 (\log t) t^{\alpha-1} (1 - t)^{\beta-1} dt &= \frac{\partial}{\partial \alpha} \int_0^1 t^{\alpha-1} (1 - t)^{\beta-1} dt \\ &= \frac{\partial}{\partial \alpha} B(\alpha, \beta) \end{aligned} \quad (12)$$

Applying this to our case with $\alpha = \alpha_y$ and $\beta = S - \alpha_y$:

$$\begin{aligned} \mathbb{E}[\log b_y] &= \frac{1}{B(\alpha_y, S - \alpha_y)} \frac{\partial}{\partial \alpha_y} B(\alpha_y, S - \alpha_y) \\ &= \frac{\partial}{\partial \alpha_y} \ln B(\alpha_y, S - \alpha_y) \end{aligned} \quad (13)$$

Step 4: Digamma function expression

Expanding the Beta function in logarithmic form:

$$\ln B(\alpha_y, S - \alpha_y) = \ln \Gamma(\alpha_y) + \ln \Gamma(S - \alpha_y) - \ln \Gamma(S) \quad (14)$$

Taking the derivative with respect to α_y :

$$\begin{aligned} \frac{\partial}{\partial \alpha_y} \ln B(\alpha_y, S - \alpha_y) &= \frac{\partial}{\partial \alpha_y} \ln \Gamma(\alpha_y) - \frac{\partial}{\partial \alpha_y} \ln \Gamma(S) \\ &= \psi(\alpha_y) - \psi(S) \end{aligned} \quad (15)$$

since S is independent of α_y .

Step 5: Final transformation

Combining these results gives:

$$\begin{aligned} \mathbb{E}_{\mathbf{b}} [-\log b_y] &= -\mathbb{E}[\log b_y] \\ &= -(\psi(\alpha_y) - \psi(S)) \\ &= \psi(S) - \psi(\alpha_y) \end{aligned} \quad (16)$$

which completes the proof.

To regularize the uncertainty estimation and prevent overconfidence in erroneous predictions, we introduce a KL divergence loss between the model’s predicted Dirichlet distribution $\text{Dir}(\alpha)$ and a non-informative prior $\text{Dir}(\beta)$:

$$\begin{aligned} \mathcal{L}_{\text{KL}} = & \log \Gamma(S) - \log \Gamma(K) - \sum_{k=1}^K \log \Gamma(\alpha_k) \\ & + \sum_{k=1}^K (\alpha_k - \beta_k) (\psi(\alpha_k) - \psi(S)) \end{aligned} \quad (17)$$

where $\Gamma(\cdot)$ is the Gamma function. The prior $\beta = (1, 1, \dots, 1)$ ($\sum_{i=1}^K \beta_i = K$) represents a “no-evidence” state, forming a uniform distribution over all classes. This setting enforces a neutral state in the absence of prior information, aligning with the nature of epistemic uncertainty. By penalizing excessive deviation from β , the loss constrains the model from overconfidence in non-true classes, enhancing generalization to ambiguous samples. The final loss function integrates the above components with baseline losses to jointly optimize feature alignment and uncertainty modeling:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{id}} + \mathcal{L}_{\text{tri}} + \lambda_1 \mathcal{L}_{\text{ce}} + \lambda_2 \mathcal{L}_{\text{KL}}. \quad (18)$$

where $\mathcal{L}_{\text{id}}$ (identity loss) and $\mathcal{L}_{\text{tri}}$ (triplet loss) serve as baseline losses for feature learning, and $\lambda_1, \lambda_2 > 0$ are two hyperparameters balancing the contributions of the probabilistic losses.

4. Experiments and Results

4.1. Datasets

4.1.1. VI-ReID Vehicle Datasets:

DN348 [16]: The training set contains 200 identities, with 9,962 daytime images and 10,022 nighttime images (50% daytime and 50% nighttime images per vehicle). For evaluation, the query set consists of 10,121 daytime images, and the gallery set includes 3,972 nighttime images, both covering 148 identities. **DNWild** [16]: Another visible-infrared cross-modal vehicle ReID dataset, containing day-night images of 2,286 identities (85,945 daytime and 54,952 nighttime images). It is used to evaluate model performance in recognizing vehicles under varying lighting conditions.

4.1.2. VI-ReID Person Datasets:

RegDB [7]: Contains visible and far-infrared images (captured by visible-light and infrared cameras) of 412 persons, totaling 8,240 images (4,120 visible and 4,120 infrared). It includes 254 female and 158 male subjects, with both frontal and rear views. **SYSU-MM01** [27]: Includes 4 RGB camera views and 2 infrared camera views (for dark environments), with 287,628 RGB images and 15,792 infrared

images of 491 distinct identities. **LLCM** [39]: Comprises 46,767 person images of 1,064 identities, where each identity is captured by at least one RGB and one IR camera. The dataset is split into training and test sets at an approximate 2:1 ratio.

4.2. Implementation Details

In our experiments, we adopted ResNet-50 pre-trained on ImageNet as the backbone, with subsequent processing following the method in DNDM [16]. We used the Stochastic Gradient Descent (SGD) optimizer, training cross-modal person and vehicle datasets for 80 epochs, respectively. Data augmentation (random horizontal flipping, random cropping, and random erasing) was applied. Input images were resized to 256×256 (vehicles) and 144×288 (persons). During training, we set 4 identities per batch, with 8 instances per identity (4 visible + 4 infrared), resulting in a batch size of 64. The margin p in Eq.(2) is set to 0.3. Hyperparameters λ_1 and λ_2 in Eq.(18) are set to 0.8 and 1, respectively. Experiments were conducted on an NVIDIA GeForce RTX 4090 GPU (24GB RAM).

4.3. Comparison with State-of-the-art Methods

4.3.1. Comparison with VI-ReID Person State-of-the-art Methods

As shown in Table 1, on the SYSU-MM01 dataset, in the “All search” scenario, CMOI achieves the highest mAP of 73.2%, outperforming DIVE and MDANet, both of which achieve 73.0%. While its Rank-1 accuracy of 75.6% is slightly lower than DIVE’s 76.9% and MDANet’s 75.8%, it remains highly competitive. In the “Indoor search” scenario, CMOI significantly outperforms all baselines, with a Rank-1 accuracy of 84.6% and an mAP of 86.3%, surpassing the second-best method, DEEN, by 4.3% and 3.0%, respectively. On the RegDB dataset, in VIS-to-IR retrieval, CMOI attains a Rank-1 accuracy of 93.5%, 1.7% higher than MDANet’s 91.8%, and an mAP of 85.3%, 0.2% higher than MAUM and DEEN’s 85.1%, ranking first. In IR-to-VIS retrieval, CMOI matches MDANet’s Rank-1 accuracy of 92.4% but achieves a higher mAP, 83.8% compared to DEEN’s 83.4%. On the LLCM dataset, CMOI leads in both retrieval directions. For VIS-to-IR, its Rank-1 accuracy of 70.8% and mAP of 55.9% outperform DEEN by 1.6% and 0.4%, respectively. In the IR-to-VIS direction, CMOI’s Rank-1 accuracy of 58.5% and mAP of 64.5% surpass DIVE by 2.7% and 1.6%, respectively.

4.3.2. Comparison with VI-ReID vehicle State-of-the-art Methods

As shown in Table 2, on the DN348 dataset, CMOI achieves the best performance in Rank-1 and Rank-5 metrics for both VIS-to-IR and IR-to-VIS tasks. In the VIS-to-IR task, CMOI’s Rank-1 (77.1%) and Rank-5 (90.9%) outper-

Table 1. Comparison of different VI-ReID methods on SYSU-MM01, RegDB, and LLCM datasets. The best and suboptimal results are marked with **red bold** and blue underline respectively.

Methods	SYSU-MM01				RegDB				LLCM			
	All Search		Indoor Search		VIS to IR		IR to VIS		VIS to IR		IR to VIS	
	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP
DDAG [34]	54.8	53.0	61.0	68.0	69.3	63.5	68.1	61.8	51.4	38.8	42.4	49.0
MMN [40]	70.6	66.9	76.2	79.6	91.6	84.1	87.5	80.5	64.0	48.5	52.5	58.9
SPOT [2]	55.4	54.1	58.5	66.3	74.2	67.6	67.5	72.4	54.8	40.8	42.8	51.0
PMT [18]	67.5	65.0	71.7	76.5	84.8	76.6	84.2	75.1	-	-	-	-
DART [31]	69.9	66.9	76.3	80.4	85.0	79.1	84.8	77.8	63.7	47.7	53.0	59.3
MAUM [17]	71.7	68.8	77.0	81.9	87.9	<u>85.1</u>	87.0	84.3	-	-	-	-
DEEN [39]	74.7	71.8	<u>80.3</u>	<u>83.3</u>	91.1	<u>85.1</u>	<u>89.5</u>	<u>83.4</u>	<u>69.2</u>	<u>55.5</u>	55.2	62.0
TopLight [37]	66.8	64.0	72.9	72.0	85.5	80.0	80.7	75.9	-	-	-	-
LCNL [32]	70.2	68.2	76.2	80.3	85.6	78.7	84.0	76.9	-	-	-	-
DIVE [6]	76.9	<u>73.0</u>	-	-	-	-	-	-	-	-	<u>55.8</u>	<u>62.9</u>
MDANet [4]	<u>75.8</u>	<u>73.0</u>	80.0	81.8	<u>91.8</u>	81.9	92.4	82.7	-	-	-	-
CMOI (Ours)	75.6	73.2	84.6	86.3	93.5	85.3	92.4	83.8	70.8	55.9	58.5	64.5

form the second-best method DNDM by 6.4% (77.1% vs. 70.7%) and 6.7% (90.9% vs. 84.2%), respectively. Although DNDM shows a slight advantage in mAP (47.5% vs. 46.9% of CMOI), the gap is negligible. In the IR-to-VIS task, CMOI ranks first with a Rank-1 of 83.2% (2.9% higher than DNDM’s 80.3%) and a Rank-5 of 94.3% (1.7% higher than DNDM’s 92.6%), while its mAP (46.4%) also exceeds DNDM’s 46.2%. On the DNWild dataset, in the VIS-to-IR task, DNDM slightly outperforms CMOI in Rank-1 (51.2% vs. 50.0%) and Rank-5 (98.7% vs. 97.7%), but CMOI achieves a better mAP (40.6% vs. 40.5%). In the IR-to-VIS task, CMOI leads in all metrics, with a 1.5% mAP improvement (41.5% vs. 40.0%) and a 1.0% Rank-1 gain (50.5% vs. 49.5%) over DNDM. Overall, CMOI demonstrates strong robustness and generalization in cross-modal vehicle retrieval.

Overall, CMOI ranks top in most key metrics across all datasets, exhibiting exceptional performance in retrieval accuracy (Rank-1) and stability (mAP), validating the effectiveness of its uncertainty-aware cross-modal interaction mechanism.

4.4. Ablation Studies

To verify the role of each core component, we conducted ablation experiments on the SYSU-MM01 (All search) and DN348 (VIS-to-IR) datasets. The experimental design involved adding each component incrementally to systematically analyze its impact on model performance (Table 3). Our Baseline (Row 1) serves as a strong, non-probabilistic model optimized only with the standard identity loss ($\mathcal{L}_{id}$) and triplet loss ($\mathcal{L}_{tri}$). This baseline does not explicitly model or utilize uncertainty.

Directly comparing our full CMOI framework (Row 4) against this non-uncertainty Baseline (Row 1) reveals the significant value of our probabilistic approach. On SYSU-MM01, the full model achieves a +1.4% gain in Rank-1 (75.6% vs 74.2%) and a +1.2% gain in mAP (73.2% vs 72.0%). The improvement is even more pronounced on DN348, where the full framework provides a substantial +8.9% leap in Rank-1 accuracy (77.1% vs 68.2%). This comparison strongly validates our central hypothesis: that explicitly modeling and fusing uncertainty is a critical strategy for mitigating the cross-modal semantic gap.

After establishing the overall value of the uncertainty framework, we further analyzed the contributions of the two specialized loss functions. Comparing the results of the second row (with only $\mathcal{L}_{KL}$) and third row (with only $\mathcal{L}_{ce}$) to the baseline (first row), each loss function improves performance when applied alone. When both losses are combined (fourth row), the model achieves a significant improvement : on SYSU-MM01, Rank-1 (75.6%) and mAP (73.2%) increase by 0.6–0.9% compared to using either loss individually ; on DN348, Rank-1 jumps from 70.0% (with either loss alone) to 77.1%, and mAP improves by 1.6–1.8%. This validates the effectiveness of dual-loss collaborative optimization: $\mathcal{L}_{ce}$ ensures alignment between the predicted distribution and ground truth , while $\mathcal{L}_{KL}$ mitigates overfitting to uncertain samples. Their combination enables the model to achieve both high accuracy and strong robustness in complex cross-modal scenarios.

These ablation experiments demonstrate that the designed uncertainty modeling mechanism and joint optimization strategy are critical for enhancing CMOI’s performance. Each component contributes complementar-

Table 2. Comparison of different VI-ReID vehicle methods on DN348 and DNWild datasets. The best and suboptimal results are marked with **red bold** and **blue underline** respectively.

Methods	DN348						DNWild					
	VIS to IR			IR to VIS			VIS to IR			IR to VIS		
	Rank-1	Rank-5	mAP	Rank-1	Rank-5	mAP	Rank-1	Rank-5	mAP	Rank-1	Rank-5	mAP
DDAG [34]	66.6	80.3	44.0	75.3	90.3	42.3	49.5	98.0	36.1	48.7	95.1	35.2
LbA [19]	68.0	82.8	44.9	76.6	89.7	42.9	47.2	93.6	26.3	41.7	87.1	25.4
AGW [36]	68.1	82.1	46.5	73.1	89.4	44.3	48.3	96.0	38.7	49.1	96.3	38.6
CAJ [35]	66.0	81.9	46.4	73.9	88.4	45.3	49.9	<u>98.1</u>	39.2	48.7	95.5	38.5
DCL [25]	67.5	81.3	44.3	78.9	92.0	42.8	49.9	97.9	34.8	47.2	94.3	34.2
PMT [18]	66.3	82.0	<u>47.0</u>	76.0	90.7	46.1	49.1	95.5	32.7	44.4	90.2	33.7
DNDM [16]	<u>70.7</u>	<u>84.2</u>	47.5	<u>80.3</u>	<u>92.6</u>	<u>46.2</u>	51.2	98.7	<u>40.5</u>	<u>49.5</u>	<u>95.5</u>	<u>40.0</u>
CMOI (Ours)	77.1	90.9	46.9	83.2	94.3	46.4	<u>50.0</u>	97.7	40.6	50.5	96.1	41.5

Table 3. Ablation Studies on the SYSU-MM01 and DN348 datasets.

Baseline	$\mathcal{L}_{ce}$	$\mathcal{L}_{KL}$	SYSU-MM01		DN348	
			Rank-1	mAP	Rank-1	mAP
✓	×	×	74.2	72.0	68.2	44.8
✓	×	✓	74.9	72.9	70.0	45.1
✓	✓	×	75.0	72.3	70.0	45.3
✓	✓	✓	75.6	73.2	77.1	46.9

ily, collectively driving improvements in cross-modal re-identification accuracy.

4.5. Other Analysis

4.5.1. Rank-list

Figure 4 shows CMOI’s visual search sequences across five VI-ReID datasets (SYSU-MM01, LLCM, RegDB, DN348, DNWild), demonstrating its cross-modal retrieval performance across query-gallery combinations (visible query-infrared gallery and infrared query-visible gallery).

4.5.2. Limitation and Future works

We acknowledge that a hyper-specialized model, optimized only for a single task, might gain an advantage on certain metrics by leveraging task-specific prior knowledge. In contrast, our work prioritizes developing a single, robust, and uncertainty-aware general framework. Future work could explore integrating more task-specific modules within our probabilistic opinion framework.

5. Conclusion

This paper proposes a novel Cross-Modal Opinion Interaction (CMOI) network. To our knowledge, this is the first work to propose a unified uncertainty-aware framework applicable to both visible-infrared person and vehicle re-identification tasks. The network effectively mitigates the cross-modal semantic gap through three stages: Firstly, addressing the limitations of traditional determinis-



Figure 4. Rank list visualization of our CMOI. Including yellow boxes indicate the queries, green boxes represent correctly retrieved images, and red boxes refer to misclassified images.

tic matching, CMOI maps evidence distributions to Dirichlet distributions to quantify cross-modal prediction uncertainty. Its output “model opinions” provide a reliable basis for decision-making, avoiding blind judgments. Secondly, a new cross-modal opinion interaction mechanism is designed to break the information barrier between visible and infrared modalities. It dynamically adjusts modal information for low-confidence predictions, fully leveraging the complementarity of multi-modal data. Finally, in the joint opinion optimization stage, the Dirichlet distribution is used as the prior distribution of class probabilities. Model parameters are optimized via cross-entropy loss (to measure the discrepancy between predictions and true categories) and Dirichlet-based regularized uncertainty loss (to regulate the model’s generalization ability for uncertain samples). Experiments verify that CMOI outperforms existing methods on five visible-infrared person and vehicle datasets, validating its effectiveness in bridging cross-modal semantic gaps.

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